



C#

Level 1

Rules of Naming Variables in *C#*



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Definition

- ◆ A variable name is the identifier used to refer to a value stored in memory.
- ◆ C# has specific rules that determine which names are valid and which are invalid.

Explanation

1. Allowed Characters

 A variable name can contain:

 Uppercase letters: A-Z

 Lowercase letters: a-z

 Digits: 0-9

 Underscore: _

 @ symbol can be used in special cases.

2. The Name Cannot Start With a Digit

◆ A variable name must start with a letter, _, or @.

3. C# Is Case-Sensitive

◆ C# treats uppercase and lowercase letters as different characters.

4. Cannot Use C# Keywords

◆ A variable name cannot normally be a C# reserved keyword.

 if

 for

 using

 class

 return

💡 Example

```
int age = 25;      // Valid  
int _age = 25;     // Valid  
int @break = 10;   // Valid  
int 101subject = 5; // Invalid
```

Note

◆ Although `_age` and `@class` are valid, they have different purposes.

▣ `_age` is commonly used by convention for private fields.

▣ `@class` is mainly used when you need an identifier that conflicts with a *C#* keyword.

⊘ Warning

- ⚠ 101subject → Invalid because it starts with a digit.
- ⚠ your name → Invalid because variable names cannot contain spaces.
- ⚠ class → Invalid because it is a *C#* keyword.
- ⚠ age and Age → They are different variables because *C#* is case-sensitive.

✦✦ Summary

- ✓ Variable names can contain letters, digits, `_`, and can use `@` in special cases.
- ✓ A variable name cannot start with a digit.
- ✓ `C#` is case-sensitive.
- ✓ Variable names cannot normally be `C#` keywords.
- ✓ `@` can be used before a keyword when necessary.

Thank you 😊